GEOMETRY · UNIT 5: SIMILARITY

Shadow Detectives

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students measure the flagpole, a tree, and a river without ever reaching them. They use shadows, mirrors, and the Side-Splitter Theorem with similar triangles, scale a school logo with a dilation, and plan a real measurement of an object on campus.

Indirect measurement is the classic application of similar triangles. Students justify AA similarity from parallel sun rays or mirror reflection angles, then explore dilations and area scale.

Standards Addressed

- G-SRT.2 Use similarity transformations
- G-SRT.3 Use transformations to establish AA similarity
- G-SRT.4 Prove theorems about triangles (side-splitter)
- G-SRT.5 Use similarity to solve problems

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch outside if possible: measure shadows. Part 1 shadow and mirror methods.
Day 2	Part 2 dilations and the Side-Splitter Theorem.
Day 3	Part 3 plan or carry out a real measurement.

Materials

- Student handout
- Measuring tapes
- Small mirrors
- Optional: a sunny day for real shadow measurements

How the Three Versions Differ

Level	What changes
CP2	Proportions set up with a guide; dilation by a whole-number scale factor; a scale drawing.
CP1	AA justification, Side-Splitter Theorem for the river, area and perimeter scale factors, and an error analysis.
Honors	Proof of AA from the mirror method, geometric mean in right triangles, and a two-method measurement with error comparison.

Watch For (Common Misconceptions)

- Setting up proportions with mismatched parts (height to shadow vs. shadow to height).
- Thinking area scales by k instead of k^2.
- Forgetting to add eye height.

Extension Ideas

- Measure a real campus object two ways and compare.
- Research Thales and the pyramids.
- Use a clinometer app and compare with trigonometry in Unit 6.

Answer Key – CP2

Part 1: Shadows & Mirrors

#	Task	Answer
1	A 5.5-foot student casts a 4-foot shadow. At the same time, the flagpole casts a 24-foot shadow. Write a proportion and find the flagpole's height.	$5.5 \div 4 = h \div 24 \rightarrow h = 33$ feet
2	A mirror lies on the ground 4 feet from a student whose eyes are 5 feet high, and 20 feet from a tree. The student sees the top of the tree in the…	$5 \div 4 = h \div 20 \rightarrow h = 25$ feet
3	Why are the two triangles in the mirror method similar?	The angles of reflection are equal and both triangles have right angles, so they are similar by AA.

Part 2: Dilations & Scale

#	Task	Answer
4	The school logo is a triangle with sides 2 cm, 3 cm, and 4 cm. It is enlarged by a scale factor of 3 for a banner. Find the new sides.	6 cm, 9 cm, 12 cm
5	How does the banner's perimeter compare with the logo's? How does its area compare?	Perimeter is 3 times larger (9 cm → 27 cm). Area is $3^2 = 9$ times larger.
6	On a campus map, 1 inch represents 40 feet. The gym is 3.5 inches long on the map. How long is the real gym?	$3.5 \times 40 = 140$ feet
7	The football field is 360 feet long. How long is it on the map?	$360 \div 40 = 9$ inches

Part 3: Plan a Measurement

#	Task	Answer
8	Choose the shadow or mirror method. Draw a labeled diagram of your plan.	Answers vary.
9	Make up realistic measurements and calculate the height.	Answers vary.
10	List two sources of error and how to reduce them.	Answers vary.

Answer Key – CP1

Part 1: Shadows & Mirrors

#	Task	Answer
1	Find the flagpole height from the shadow data. State the similarity criterion and why it applies.	h = 33 ft; AA (both triangles have a right angle and the same sun angle).
2	Mirror method: eye height 5 ft, 4 ft to the mirror, 20 ft from the mirror to a tree. Find the tree's height. Then find it again if the eye height…	h = 25 ft; with 5.25 ft: h = 5.25 × 20 ÷ 4 = 26.25 ft, a difference of 1.25 ft.
3	A student 6 ft tall casts a 7.5-ft shadow while a tree casts a 30-ft shadow, measured 2 hours later. Explain why the result may be wrong.	The sun angle changes over two hours, so the triangles are no longer similar.

Part 2: River & Logo

#	Task	Answer
4	To measure a river, a surveyor uses parallel segments as shown. Find the river width w.	$18 \div 24 = w \div (w + 12) \rightarrow 18w + 216 = 24w \rightarrow w = 36$ m
5	In $\triangle ABC$, DE ∥ BC with D on AB and E on AC. AD = 12, DB = 8, and AE = 15. Find EC.	$12 \div 8 = 15 \div EC \rightarrow EC = 10$
6	The logo (area 2.9 cm^2) is enlarged by a scale factor of 3. Find the banner's area. What scale factor would make the area 100 times larger?	$2.9 \times 9 = 26.1$ cm^2; a scale factor of 10 ($10^2 = 100$).

Part 3: Two-Method Measurement

#	Task	Answer
7	Draw diagrams for the shadow method and the mirror method.	Answers vary.
8	Use realistic made-up (or real) data to compute the height both ways.	Answers vary.
9	Compare the results and explain which method is more reliable.	Answers vary.

Answer Key – Honors

Part 1: Prove & Measure

#	Task	Answer
1	Prove that the two triangles in the mirror method are similar (the angle of incidence equals the angle of reflection).	Both triangles have a right angle with the ground, and the angles at the mirror are congruent (incidence = reflection), so they are similar by AA.
2	Mirror data: eye height e, distance to mirror a, mirror to tree b. Write a formula for the tree height h. Use e = 5, a = 4, b = 20.	h = e · b ÷ a; h = 5 · 20 ÷ 4 = 25 ft
3	If each distance can be off by ±0.25 ft, find the smallest and largest possible tree heights.	Smallest: 4.75 · 19.75 ÷ 4.25 ≈ 22.07 ft. Largest: 5.25 · 20.25 ÷ 3.75 = 28.35 ft.

Part 2: Side-Splitters & Geometric Means

#	Task	Answer
4	Find the river width w in the diagram.	w = 36 m
5	Prove the Side-Splitter Theorem: if DE ∥ BC in △ABC, then AD ÷ DB = AE ÷ EC.	DE ∥ BC gives corresponding angles ≅, so △ADE ~ △ABC (AA). Then AD ÷ AB = AE ÷ AC; rewrite AB = AD + DB and AC = AE + EC and simplify to AD ÷ DB = AE ÷ EC.
6	In a right triangle, the altitude to the hypotenuse divides it into segments of 4 and 9. Find the altitude and both legs.	Altitude = √(4 · 9) = 6; legs √(4 · 13) = √52 = 2√13 ≈ 7.21 and √(9 · 13) = √117 = 3√13 ≈ 10.82

Part 3: Field Report

#	Task	Answer
7	Give diagrams, data, and both calculations.	Answers vary.
8	Use your error ranges to decide whether the two results agree.	Answers vary.
9	Write the report with a recommendation for the yearbook.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Measurement plan uses a valid similarity method with a clear diagram and realistic data. Goes beyond the minimum with a creative, realistic choice.	Measurement plan uses a valid similarity method with a clear diagram and realistic data.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes diagrams, the similar-triangle justification, proportions solved with units, and an error analysis. Polished, organized, and visually engaging.	Report includes diagrams, the similar-triangle justification, proportions solved with units, and an error analysis. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.